P E M

Student Number

2025 **TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION**

Physics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Draw diagrams using pencil
- Calculators approved by NESA may be used
- A data sheet, formulae sheet and Periodic Table are provided

Total marks: 100

Section I – 20 marks (pages 2–8)

- Attempt Questions 1–20
- Allow about 35 minutes for this section

Section II – 80 marks (pages 9–22)

- Attempt Questions 21–32
- Allow about 2 hours and 25 minutes for this section

Section I – 20 marks
Attempt Questions 1-20
Allow about 35 minutes for this section.

Use the multiple-choice answer sheet for Questions 1-20

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1 A car travels around a flat circular track at constant speed. Which of the following best describes the net force acting on the car?

(A) Tangent to the circle in the direction of motion.

(B) Directed outward away from the centre.

(C) Directed inward toward the centre.

(D) Zero.

2 Which type of radioactive decay results in the emission of a helium nucleus?

(A) Alpha decay.

(B) Beta-minus decay.

(C) Beta-plus decay.

(D) Gamma decay.

3 Which condition is most critical for sustaining nuclear fusion in a star's core?

(A) Low pressure.

(B) Presence of hydrogen or helium.

(C) Presence of heavy elements.

(D) High pressure and temperature.

4 Which of the following correctly describes the cause of electromagnetic induction?

(A) A changing electric field produces a changing magnetic field.

(B) A magnetic field induces a current when it changes relative to a conductor.

(C) A static magnetic field creates a static current.

(D) Electric charges moving at constant velocity produce an induced current.

5 Which form of electromagnetic radiation has the highest energy per photon?

(A) X-rays

(B) Ultraviolet

(C) Infrared

(D) Microwaves

6 Which change would result in the greatest increase in the gravitational force between two objects?

(A) Double the distance between the objects.

(B) Double the mass of one object.

(C) Double the mass of both objects.

(D) Halve the mass of both objects.

7 Which of the following observations could not be explained by Rutherford’s atomic model?

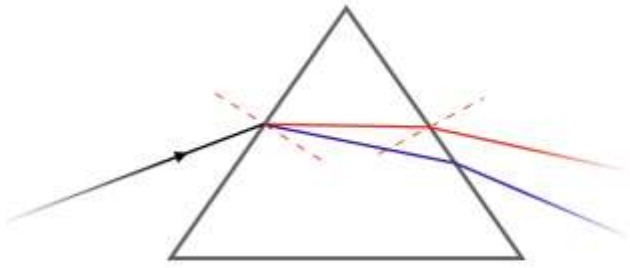
(A) Most alpha particles passed through gold foil.

(B) Some alpha particles were deflected at large angles.

(C) Atoms are mostly empty space.

(D) Atoms emit discrete spectral lines.

8 When white light passes through a prism, red light bends less than violet light.



This is because:

(A) Violet light has a shorter wavelength and refracts more.

(B) Red light has a higher frequency than violet light.

(C) Red light travels slower in glass than violet light.

(D) The energy of red photons is greater than that of violet photons.

9 As a satellite in an elliptical orbit moves closer to Earth, what happens to its kinetic and potential energy?

(A) Both kinetic and potential energy increase.

(B) Kinetic energy increases, potential energy decreases.

(C) Both kinetic and potential energy decrease.

(D) Kinetic energy decreases, potential energy increases.

10 Which combination of quarks correctly represents a neutron?

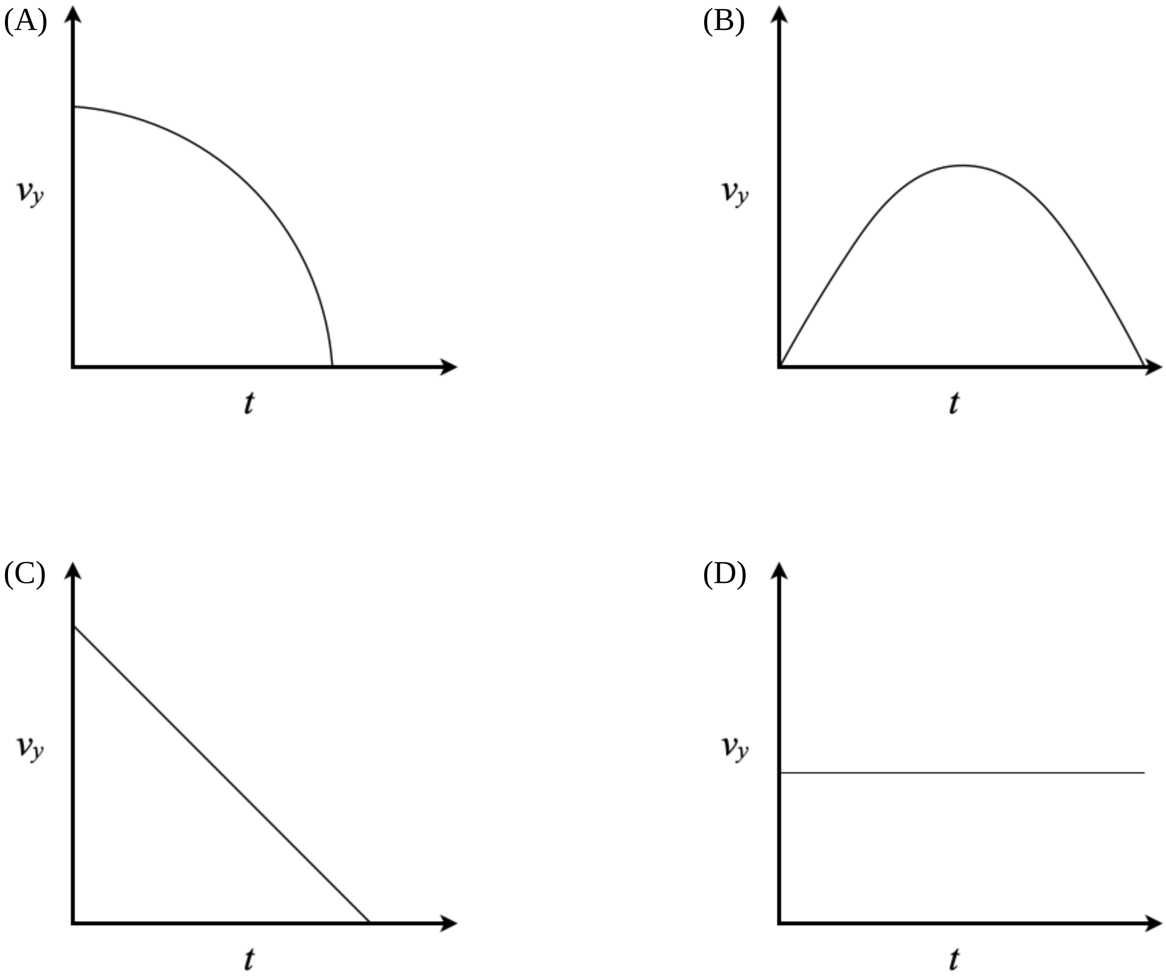
(A) Down, Down, Up

(B) Up, Up, Down

(C) Up, Down, Strange

(D) Charm, Down, Down

11 A projectile is launched horizontally from a cliff. Which graph best represents the vertical velocity of the projectile over time?



12 In an electromagnetic wave travelling through a vacuum, the electric field and magnetic field are:

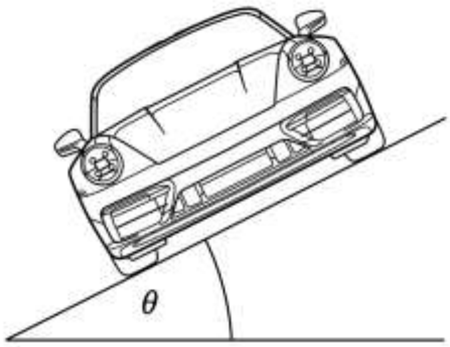
(A) Parallel and constant in magnitude.

(B) Parallel but oscillating in opposite directions.

(C) Perpendicular to each other and to the direction of wave travel.

(D) Randomly aligned but always oscillating.

13 A diagram shows a car on a frictionless banked curve.



Which of the following factors primarily determines the banking angle needed to negotiate the curve safely?

(A) The mass of the car.

(B) The speed of the car.

(C) The shape of the tires.

(D) The type of surface material.

14 In an H–R diagram, where would a star with low temperature but high luminosity be found?

(A) Main sequence.

(B) White dwarf region.

(C) Neutron star region.

(D) Red giant region.

15 A nucleus of ^{238}U undergoes alpha decay. What are the atomic and mass numbers of the daughter nucleus?

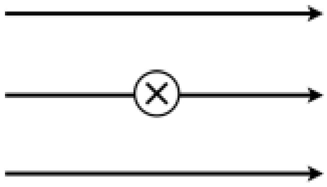
(A) 90 and 236.

(B) 92 and 234.

(C) 90 and 234.

(D) 88 and 236.

16 A straight wire carries a current into the page in a uniform magnetic field directed to the right as shown below.



What is the direction of the force on the wire?

(A) Into the page.

(B) Out of the page.

(C) Upward.

(D) Downward.

17 A transformer has a primary voltage of 500 V and a secondary voltage of 100 V. The primary current is 1.5 A. If the transformer is 100% efficient, what is the secondary current?

(A) 0.3 A

(B) 1.5 A

(C) 2.5 A

(D) 7.5 A

18 A proton moves at 2.0×10^{6} m s^{-1} perpendicular to a magnetic field and experiences a force of 3.2×10^{-14} N.

What is the strength of the magnetic field?

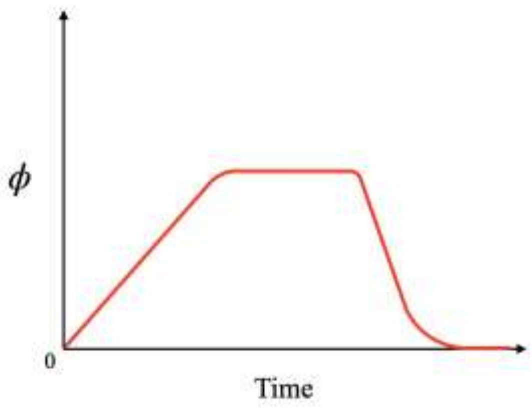
(A) 0.08 T

(B) 0.10 T

(C) 0.24 T

(D) 0.32 T

19 A graph shows the magnetic flux through a coil as a function of time.



Which time interval shows the largest induced emf?

(A) When flux is constant.

(B) When flux increases gradually.

(C) When flux oscillates about zero.

(D) When flux decreases rapidly.

20 A beam of light passes through a diffraction grating. Compared to passing through a single slit, the diffraction grating causes:

(A) Broader and less distinct fringes.

(B) Narrower and more distinct fringes.

(C) No interference at all.

(D) Complete polarisation of the light.

Sections II – 80 marks
Attempt questions 21-32
Allow about 2 hour and 25 minutes for this section.

Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

Show all relevant working in questions involving calculations.

Question 21 (5 marks)

Modern electric vehicles use regenerative braking systems.

(a) Explain how Lenz's Law is demonstrated during regenerative braking. **2**

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(b) Describe how the regenerative braking force changes as the car slows down. **3**

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Question 22 (6 marks)

A lunar rover is navigating the surface of the Moon, where the gravitational acceleration is 1.62 m s^{-2}. The rover has a mass of 200 kg.

It drives over a circular hill of radius 30 m at a constant speed of 8.0 m s^{-1}.

(a) Calculate the normal force acting on the rover at the top of the hill. **3**

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(b) Explain how and why the normal force would differ if the same hill were on Earth instead of the Moon. **3**

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Question 23 (5 marks)

A sample of a radioactive isotope has an initial activity of 6400 counts per minute. After 12 hours, its activity has decreased to 800 counts per minute.

(a) Determine the half-life of the isotope. **3**

(b) How many atoms remain compared to the original number after 36 hours? **2**

Question 24 (6 marks)

Projectile A is launched at 60˚ from the horizontal. Projectile B is launched at 45˚ from the horizontal when projectile A is halfway through its flight.

Determine the ratio of projectile B's initial velocity to projectile A's in order for them to collide on their flight paths.

Question 25 (9 marks)

A space station orbits Earth at an altitude of 400 km.

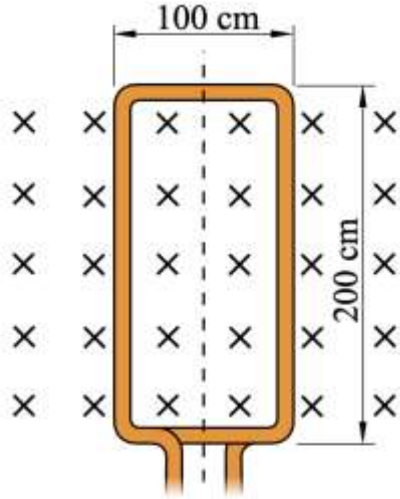
(a) Calculate the orbital speed of the station. **3**

(b) If the station wants to move to a higher circular orbit at an altitude of 800 km, describe and explain the energy changes involved. **3**

(c) Outline why satellites in higher orbits require less speed to maintain orbit than satellites in lower orbits. Support your answer with reference to gravitational and centripetal forces. **3**

Question 26 (5 marks)

A rectangular coil with 20 turns is placed in a uniform magnetic field of 0.30 T, as shown below. The coil is rotated at a constant angular velocity of 150 rpm about an axis perpendicular to the magnetic field.



(a) The diagram above shows the coil at a particular instant in its rotation. Using this diagram, explain why the induced emf is zero at that instant. **2**

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(b) Determine the average induced emf generated in the coil after it has rotated through 90˚ from its shown position. **3**

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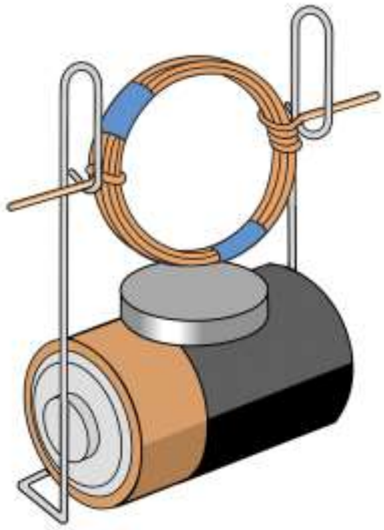
Question 27 (7 marks)

(a) Explain how the photoelectric effect provides evidence for the particle nature of light. **3**

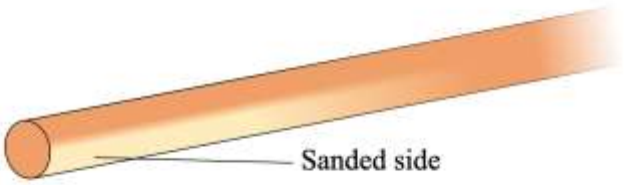
(b) Describe how varying the frequency and intensity of incident light would affect the observed current. Support your response with reference to classical wave theory vs quantum theory. **4**

Question 28 (10 marks)

Students, in class, constructed a simple electric motor with an insulated copper coil, paperclips, magnet, and a battery as shown below.



The teacher instructed the students to only sand one side of the copper wire, as shown below, where it contacted the paperclips.



(a) Explain and justify the sanding instruction. **4**

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Question 28 continues on page 17

Question 28 (continued)

(b) Identify which part of a DC motor works in the same way as the half-sanded wire section? **1**

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(c) One student did not follow the instruction properly and sanded both sides of the wire. **2**

Describe what will happen to this student's coil when the power is turned on.

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(d) When the students measured the currents through their motors the student who did not sand their wire correctly noticed their current was significantly higher than the other students. Explain this observation. **3**

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Question 29 (6 marks)

In 1801, Thomas Young demonstrated the interference of light in his famous double-slit experiment.

(a) Describe how the observed interference pattern supports the wave model of light. **2**

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(b) Discuss how later experiments with single photons challenge this model, and explain how quantum theory resolves the apparent contradiction. **4**

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Question 30 (6 marks)

Explain how Maxwell's equations predicted the existence of electromagnetic waves and outline how this prediction was later verified experimentally.

Question 31 (7 marks)

During nuclear fusion in stars, two deuterium nuclei combine to form a helium-3 nucleus and a neutron, according to the reaction:

$$^{2}_{1}\mathrm{H} + ^{2}_{1}\mathrm{H} \rightarrow ^{3}_{2}\mathrm{He} + ^{1}_{0}\mathrm{n}$$

The masses are:

- $^{2}_{1}\mathrm{H} = 2.0141$ u
- $^{3}_{2}\mathrm{He} = 3.0160$ u
- $^{1}_{0}\mathrm{n} = 1.0087$ u

(a) Calculate the mass defect for the fusion reaction. **3**

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(b) Determine the energy released in joules by a single fusion event. **2**

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(c) Calculate the total energy released if 1.5×10^{25} deuterium nuclei undergo fusion. **2**

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Question 32 (8 marks)

Discuss the role of evidence from stellar spectroscopy and nuclear fusion in the Sun in supporting current models of nucleosynthesis.

END OF EXAM

PEM

2025 Trial HSC Physics
Marking Guidelines

Section I

Multiple-choice Answer Key

Question	Answer
1	C
2	A
3	D
4	B
5	A
6	C
7	D
8	A
9	B
10	A
11	C
12	C
13	B
14	D
15	C
16	D
17	D
18	B
19	D
20	B

Section II

Question 21 (a)

Criteria	Marks
Correctly explains how Lenz's Law applies, making specific reference to the induced current opposing motion and regenerating braking.	2
Provides a basic description of either Lenz's Law or regenerative braking without clearly linking them.	1

Sample answer:

During regenerative braking, the moving vehicle's wheels drive an electric motor in reverse, acting as a generator. According to Lenz's Law, the induced current creates a magnetic field that opposes the motion of the wheels, slowing the vehicle while converting kinetic energy into electrical energy.

Question 21 (b)

Criteria	Marks
Clearly describes how the braking force decreases as the vehicle slows down, relating to reduction in induced current and electromotive force (EMF).	3
Provides a basic description that braking force decreases, with partial explanation.	2
Identifies that braking force changes but with little or no explanation.	1

Sample answer:

As the car slows down, the wheels turn more slowly, reducing the rate of change of magnetic flux through the motor. This results in a smaller induced EMF and a weaker induced current.
Consequently, the regenerative braking force decreases as the car slows down.

Question 22 (a)

Criteria	Marks
Correctly calculates the normal force, showing correct working and substitution.	3
Shows correct working with a minor error in substitution or final calculation.	2
Identifies the correct relevant formula(s) or attempts correct working.	1

Sample answer:

$-F_c = F_g + F_N$

$F_N = -\frac{mv^2}{r} - F_g$

$F_N = -\frac{200\times 8^2}{30} - (200 \times -1.62)$

$F_N = -103$ N

Because the answer is negative, the rover has left the surface.

$\therefore F_N = 0$

Question 22 (b)

Criteria	Marks
Clearly explains how and why the normal force would be greater on Earth, making specific reference to gravitational acceleration and centripetal force requirements.	3
Provides a general explanation of either how or why the normal force changes, with partial reference to gravity or motion.	2
Identifies that the normal force would be different but gives a vague or incomplete reason.	1

Sample answer:

On Earth, the gravitational acceleration is much larger (9.8 m s^{-2} compared to 1.62 m s^{-2} on the Moon). This means the rover's weight would be much greater on Earth. Since the normal force is the difference between gravitational force and the required centripetal force, $F_N = -\frac{mv^2}{r} - F_g$, the larger gravitational force on Earth would result in a larger normal force compared to the Moon.
Thus, the rover would maintain contact with the surface more easily on Earth, and the normal force would not fall to zero as quickly.

Question 23 (a)

Criteria	Marks
Correctly determines the number of half-lives, calculates the half-life with correct working.	3
Attempts correct working but contains a minor error in calculation.	2
Identifies the basic method or formula to find half-life.	1

Sample answer:

$6400 \rightarrow 3200 \rightarrow 1600 \rightarrow 800$

$n = 3$

$n = \frac{t}{t_{1/2}}$

$t_{1/2} = \frac{t}{n}$

$t_{1/2} = \frac{12}{3}$

$t_{1/2} = 4$ hrs

Question 23 (b)

Criteria	Marks
Correctly calculates the remaining fraction of atoms after 36 hours, showing clear working.	2
Attempts correct working or correctly identifies the number of half-lives.	1

Sample answer:

$n = \frac{t}{t_{1/2}}$

$n = \frac{36}{4}$

$n = 9$

$N_t = N_0 \left(\frac{1}{2}\right)^n$

$N_t = N_0 \left(\frac{1}{2}\right)^9$

$N_t = N_0 \left(\frac{1}{512}\right)$

$\frac{1}{512}$ of the original atoms remain after 36 hrs.

Question 24

Criteria	Marks
Correctly determines ratio of velocities showing full working	6
Determines a value for the velocity ratio with minor error.	5
Calculates horizontal displacement for both projectiles OR sets displacements equal for collision condition. Shows algebraic progress toward velocity ratio.	4
Correctly calculates horizontal velocity component of **either** projectile using time of flight expressions.	3
Correctly determines an expression for time of flight for either projectile A or B using appropriate trigonometric substitution.	2
Identifies the need to compare horizontal displacements or discusses timing of the two projectiles. May correctly write or use a formula for time of flight.	1

Sample answer:

Solving for collision when projectile A lands → $t = total\ time$

Projectile A.

$t = \frac{2u\sin\theta}{g}$

$t_A = \frac{2u_A \sin 60}{g}$

$t_A = \frac{u_A\sqrt{3}}{g}$

$u_{Ax} = u_A \cos 60$

$u_{Ax} = \frac{u_A}{2}$

$\Delta x_A = u_{Ax} \times t_A$

$\Delta x_A = \frac{u_A}{2} \times \frac{u_A\sqrt{3}}{g}$

$\Delta x_A = \frac{{u_A}^2\sqrt{3}}{2g}$

Projectile B.

Proj. B has a flight time half that of proj. A.

$t_B = \frac{u_A\sqrt{3}}{2g}$

$u_{Bx} = u_B \cos 45$

$u_{Bx} = \frac{u_B\sqrt{2}}{2}$

$\Delta x_B = u_{Bx} \times t_B$

$\Delta x_B = \frac{u_B\sqrt{2}}{2} \times \frac{u_A\sqrt{3}}{2g}$

$\Delta x_B = \frac{u_A u_B\sqrt{6}}{4g}$

Collision occurs at $\Delta x_A = \Delta x_B$

$$\frac{{u_A}^2\sqrt{3}}{2g} = \frac{u_A u_B \sqrt{6}}{4g}$$

$$2{u_A}^2\sqrt{3} = u_A u_B \sqrt{6}$$

$$2u_A\sqrt{3} = u_B\sqrt{6}$$

$$\frac{u_B}{u_A} = \frac{2\sqrt{3}}{\sqrt{6}}$$

$$\frac{u_B}{u_A} = \sqrt{2}$$

Question 25 (a)

Criteria	Marks
Correctly calculates the orbital speed with full correct working and substitution.	3
Shows correct method and working but with a minor error.	2
Identifies the correct formula or method to solve.	1

Sample answer:

$$v = \sqrt{\frac{GM}{r}}$$

$$v = \sqrt{\frac{6.67\times10^{-11}\times6\times10^{24}}{4\times10^5+6.371\times10^6}}$$

$v = 7690$ m s^{-1}

Question 25 (b)

Criteria	Marks
Clearly describes and explains the increase in gravitational potential energy and corresponding change in kinetic energy.	3
Describes the energy changes but with limited explanation of the relationship between energy types.	2
Identifies an energy change but with no clear explanation.	1

Sample answer:

To move to a higher orbit, the station must gain gravitational potential energy because it moves farther from Earth's centre. At the same time, its kinetic energy decreases because the orbital speed required is lower at higher altitudes. Thus, there is a trade-off: some kinetic energy is converted into gravitational potential energy.

Question 25 (c)

Criteria	Marks
Correctly outlines the need for lower orbital speed, linking gravitational force and required centripetal force.	3
Provides a general outline but limited connection between forces and orbital speed.	2
Identifies that less speed is needed but provides little or no reasoning.	1

Sample answer:

Satellites in higher orbits are farther from Earth's centre, so the gravitational force acting on them is weaker. Since the gravitational force provides the centripetal force necessary to maintain orbit, a weaker force means a lower required orbital speed. Thus, satellites in higher orbits move more slowly to stay in stable circular motion.

Question 26 (a)

Criteria	Marks
Clearly explains that no change in magnetic flux occurs when the coil plane is parallel to the magnetic field direction.	2
Identifies that the emf is zero but gives limited or incomplete explanation.	1

Sample answer:

At this instant, the plane of the coil is perpendicular to the magnetic field, so the magnetic flux is at a maximum. However, the rate of change of flux is zero at this point in the rotation, so no emf is induced.

Question 26 (b)

Criteria	Marks
Correctly calculates the induced emf using Faraday's Law, correctly finding flux change and time.	3
Shows correct working using Faraday's Law with a minor error in substitution or arithmetic.	2
Identifies appropriate method to find emf from flux change over time.	1

Sample answer:

$\Phi = BA = 0.3 \times (1.0 \times 2.0)$

$\Phi = 0.6$ Wb

$T = \frac{60}{150} = 0.4$ s per revolution

$\Delta t = \frac{0.4}{4} = 0.1$ s

$\varepsilon = N\frac{\Delta\Phi}{\Delta t}$

$\varepsilon = 20\frac{0.6}{0.1}$

$\varepsilon = 120$ V

Question 27 (a)

Criteria	Marks
Clearly explains how the photoelectric effect demonstrates light behaving as discrete particles (photons) and links to threshold frequency.	3
Identifies particle nature or threshold frequency but provides incomplete explanation.	2
States that photoelectric effect shows particle-like behaviour without clear support.	1

Sample answer:

In the photoelectric effect, electrons are only emitted from a metal surface if the incident light has a frequency above a certain threshold, regardless of its intensity. This suggests that light energy is delivered in discrete packets (photons) rather than as a continuous wave. Only photons with enough energy ($E=hf$) can transfer energy to electrons, supporting the particle model of light.

Question 27 (b)

Criteria	Marks
Correctly describes the effects of varying both frequency and intensity on the photoelectric current, contrasting classical and quantum predictions.	4
Describes most effects correctly but with minor omission or limited contrast between theories.	3
Provides basic description of one or both effects without clear link to classical vs quantum theory.	2
Identifies some change to current with vague or incomplete explanation.	1

Sample answer:

According to quantum theory, increasing the frequency of light increases the energy of individual photons. If the frequency is below the threshold, no current is observed. If it is above the threshold, increasing frequency does not increase the current, but increases the kinetic energy of emitted electrons. Increasing the intensity increases the number of photons striking the surface per second, so the current increases (more electrons are emitted).

Classical wave theory incorrectly predicted that increasing intensity (regardless of frequency) should always increase current, and that there should be no threshold frequency.
The experimental results supported quantum theory instead.

Question 28 (a)

Criteria	Marks
Clearly explains that sanding is required to expose the conductive copper *and* justifies how half-sanding allows controlled current flow for continuous rotation.	4
Identifies the need to expose the copper *and* attempts to link half-sanding to motor operation.	3
Identifies either exposing copper **or** intermittent current flow, with limited or no clear justification.	2
Mentions sanding affecting operation but provides minimal explanation.	1

Sample answer:

The wire is initially coated with an insulating layer, so sanding is required to expose the conductive copper underneath to allow current to flow into the coil. Sanding only one side of the wire ensures that current only flows during part of the rotation. When the bare copper side contacts the paperclip, current flows, generating a magnetic field and torque. When the insulated side contacts the paperclip, current is interrupted, allowing the coil to continue spinning without being pulled backward. This intermittent current flow is essential for maintaining continuous rotation in one direction.

Question 28 (b)

Criteria	Marks
Correctly identifies the commutator.	1

Sample answer:

The commutator.

Question 28 (c)

Criteria	Marks
Correctly describes that the coil will oscillate back and forth or not spin properly.	2
Identifies that the coil will not spin correctly but with limited description.	1

Sample answer:

If both sides are sanded, current will flow continuously regardless of the coil's orientation.
This causes the magnetic torque to reverse direction halfway through each rotation, making the coil oscillate back and forth instead of spinning continuously.

Question 28 (d)

Criteria	Marks
Correctly explains that the absence of back emf leads to a higher current because the supply voltage is not opposed.	3
Identifies reduced back emf or higher current but with limited explanation.	2
States current would be higher but without clear or correct reasoning.	1

Sample answer:

In a normal motor, the spinning coil generates a back emf that opposes the supply voltage, reducing the net current through the circuit.
Since the student's coil was not rotating properly, little or no back emf was generated.
Without back emf to oppose the battery voltage, the full voltage was applied across the low-resistance coil, resulting in a much higher current compared to other students.

Question 29 (a)

Criteria	Marks
Correctly describes how constructive and destructive interference leads to a pattern of bright and dark fringes, supporting the wave model.	2
Identifies that a pattern forms but provides limited or vague explanation of interference.	1

Sample answer:

The observed pattern of bright and dark fringes is due to constructive and destructive interference of light waves from the two slits. This supports the wave model because interference is a characteristic behaviour of waves.

Question 29 (b)

Criteria	Marks
Discusses the challenge posed by single-photon experiments and clearly explains how quantum theory (wave-particle duality) resolves the contradiction.	4
Identifies the contradiction and gives some explanation of quantum theory with minor omission or unclear discussion.	3
Describes either the contradiction or quantum explanation with limited connection between them.	2
States that single photons challenge the wave model but gives minimal explanation.	1

Sample answer:

When single photons are sent through the double slits, each photon appears to interfere with itself over time, producing an interference pattern even though photons arrive one at a time.
This challenges the classical wave model, which predicts that interference requires two waves interacting simultaneously.

Quantum theory resolves this by proposing wave-particle duality: each photon behaves as a probability wave, spreading out and interfering with itself.
The interference pattern builds up over time as more photons are detected, showing both wave-like and particle-like properties.

Question 30

Criteria	Marks
Clearly explains how Maxwell’s equations predict electromagnetic waves through changing electric and magnetic fields, and outlines Hertz’s experimental verification.	6
Explains electromagnetic wave prediction and mentions experimental verification but with some minor omissions or lack of clarity.	5
Correctly explains the prediction or correctly outlines experimental verification, but only partially addresses the other.	4
Describes either the prediction or the experimental verification with limited explanation.	3
Identifies that Maxwell predicted electromagnetic waves or that Hertz verified them, with little detail.	2
Makes a minimal relevant statement about electromagnetic waves.	1

Sample answer:

Maxwell's equations describe how changing electric fields produce magnetic fields, and changing magnetic fields produce electric fields. He predicted that a self-sustaining electromagnetic wave could propagate through space as oscillating electric and magnetic fields reinforcing each other.
Maxwell calculated that these waves would travel at a speed close to the known speed of light, suggesting that light itself is an electromagnetic wave.

This prediction was later verified by Heinrich Hertz in 1887. Hertz generated radio waves using an oscillating electric spark and detected them with a simple loop antenna. He observed that the waves could reflect, refract, polarise, interfere, and diffract, confirming that they behaved like light and other known electromagnetic waves, thus validating Maxwell's prediction.

Question 31 (a)

Criteria	Marks
Correctly calculates the mass defect with clear working and correct substitution.	3
Shows correct method but with minor calculation or substitution error.	2
Attempts to find mass defect using masses given.	1

Sample answer:

$\Delta m = (m\ of\ reactants) - (m\ of\ products)$

$\Delta m = (2 \times 2.0141) - (3.0160 + 1.0087)$

$\Delta m = 0.0035$ u

Question 31 (b)

Criteria	Marks
Correctly calculates energy released in joules using mass-energy equivalence.	2
Attempts correct calculation but with a minor error.	1

Sample answer:

$E = mc^2$

$E = (0.0035 \times 1.661 \times 10^{-27}) \times (3 \times 10^8)^2$

$E = 5.23 \times 10^{-13}$ J

Question 31 (c)

Criteria	Marks
Correctly calculates the number of fusion events and total energy released.	2
Attempts correct method but with minor error in calculation.	1

Sample answer:

$Fusion\ events = \frac{1.5\times10^{25}}{2} = 7.5 \times 10^{24}$

$E_{total} = 7.5 \times 10^{24} \times 5.23 \times 10^{-13}$

$E_{total} = 3.92 \times 10^{12}$ J

Question 32

Criteria	Marks
Provides a comprehensive and detailed discussion linking evidence from both stellar spectroscopy and nuclear fusion to the development of nucleosynthesis models. Demonstrates clear understanding of how observations support theoretical models.	7-8
Provides a clear discussion of evidence from both spectroscopy and nuclear fusion, with some links to nucleosynthesis models. May have minor omissions or less depth in connecting evidence to theory.	5-6
Describes evidence from spectroscopy or nuclear fusion with limited link to nucleosynthesis. May describe observations without clearly explaining how they support the model.	3-4
Identifies basic features of spectroscopy or fusion but provides little or no connection to nucleosynthesis models. Discussion is minimal or largely descriptive.	1-2

Sample answer:

Stellar spectroscopy provides direct evidence of the elements present in stars by analysing the absorption and emission spectra. Different elements produce characteristic spectral lines, and the presence of hydrogen, helium, and heavier elements in various stars supports the idea that stars undergo processes that change their elemental composition over time.

Observations show that older stars have lower metallicity (fewer elements heavier than helium), suggesting that heavier elements are produced inside stars and dispersed over successive generations. Nuclear fusion in the Sun provides a model for how elements are created. In the Sun's core, hydrogen nuclei fuse through a series of reactions (mainly the proton-proton chain) to form helium, releasing energy. This confirms that lighter elements can combine to form heavier elements under extreme temperatures and pressures.

The Sun's energy output matches the predicted fusion rates based on models, providing strong support for nuclear fusion as the mechanism behind stellar energy production and nucleosynthesis.
Together, stellar spectroscopy and fusion processes observed in the Sun validate the current model of nucleosynthesis, where stars are the primary sites of element formation, from hydrogen up to iron, and heavier elements are produced in supernovae and other high-energy events.

Physics

2025 Trial HSC Examination Mapping Grid

Section I

Question	Marks	Outcome
1	1	PH12-12
2	1	PH12-15
3	1	PH12-15
4	1	PH12-13
5	1	PH12-14
6	1	PH12-12
7	1	PH12-15
8	1	PH12-14
9	1	PH12-12
10	1	PH12-15
11	1	PH12-12
12	1	PH12-13
13	1	PH12-12
14	1	PH12-15
15	1	PH12-15
16	1	PH12-13
17	1	PH12-13
18	1	PH12-13
19	1	PH12-13
20	1	PH12-14

Section II

Question	Marks	Outcome
21 (a)	2	PH12-13
21 (b)	3	PH12-13
22 (a)	3	PH12-12
22 (b)	3	PH12-12
23 (a)	3	PH12-15
23 (b)	2	PH12-15

Question	Marks	Outcome
24	6	PH12-12
25 (a)	3	PH12-12
25 (b)	3	PH12-12
25 (c)	3	PH12-12
26 (a)	2	PH12-13
26 (b)	3	PH12-13
27 (a)	3	PH12-14
27 (b)	4	PH12-14
28 (a)	4	PH12-13
28 (b)	1	PH12-13
28 (c)	2	PH12-13
28 (d)	3	PH12-13
29 (a)	2	PH12-14
29 (b)	4	PH12-14
30	6	PH12-13
31 (a)	3	PH12-15
31 (b)	2	PH12-15
31 (c)	2	PH12-15
32	8	PH12-15